3 Methodology

In our experiment, a *reasoning problem* is given by a list of views (the *premises*), and the problem is to answer the question "what if anything follows?". Our method generates reasoning problems where the ETR-predicted answer is a logical fallacy.

3.1 Data processing

Table 1: Original bank of reasoning problems.

Problem	Symbolic representation	Natural language example
Modus ponens	$\{R(x)\}^{\{Q(x)\}}$ $\{Q(x)\}$ $\therefore \{R(x)\}$	If it rains, it's wet. It rains. Therefore, it's wet.
Modus tollens	$\{R(x)\}^{\{Q(x)\}}$ $\{\overline{R(x)}\}$ $\therefore \{\overline{Q(x)}\}$	If the switch is on, then the lamp is lit. The lamp is not lit. Therefore, the switch is not on.
Quantified modus ponens	$\forall x\{R(x)\}^{\{Q(x)\}}$ $\forall x\{Q(x)\}^{\{P(x)\}}$ $\therefore \forall x\{R(x)\}^{\{P(x)\}}$	All mammals have lungs. All dogs are mammals. Therefore, all dogs have lungs.
Disjunction fallacy	$\{Q(x)R(x), S(x)T(x)\}$ $\{Q(x)\}$ $\therefore \{R(x)\}$	She is quiet and clever, or tall and athletic. She is quiet. Therefore, she is clever.

We source our original bank of reasoning problems from examples presented in the *Reason and Inquiry* text [10]. These include basic "template" reasoning problems like modus ponens, modus tollens, and the disjunction fallacy as originally presented in Walsh and Johnson-Laird [6]. Table 1 presents the complete list of original problems. Not all reasoning problems take the form of valid syllogistic inferences; that is, not all conclusions from Erotetic Theory inference deductively follow from the premises.

To avoid complications in the mapping to natural language, we restrict to views with only *monadic* (one-place) predicates. The corresponding monadic first-order logic is known to be less expressive than full first-order logic, but still allows for a rich set of reasoning problems. Our method could easily be extended to include polyadic predicates.

3.2 Generative pipeline

We define a collection of mutation functions that give slight modifications to ETR view objects. These functions serve to expand our bank of reasoning problems by modifying existing problems by single premises. Table 2 presents the complete list of possible mutation rules.

Generating a new problem proceeds by iterative building a list of views (the premises in the problem). A new view is selected at random from the original problem bank and subjected to a random number of random mutation rules, it is added to the list if ETR predicts the problem admits a non-trivial answer. New views are added until all of the following stopping conditions are met. (1) The problem is the right size, meaning that summing the number of atoms in each view gives between 4 and 11. If 11 is exceeded, backtracking is used. (2) The ETR-predicted conclusion contains a single alternative (no disjunctions). (3) The ETR-predicted conclusion is a logical fallacy. To mitigate concerns about dataset-specific effects, we intentionally generated problems across diverse domains and structures. The PyETR-based generation ensures our problems test reasoning capacities rather than memorization of training examples.

Dataset curation We initially generated 400 problems. Pre-analysis integrity checks identified 17 items whose fallacy status did not meet the final specification, so we excluded them prior to all analyses, yielding 383 problems. Given the modest reduction in sample size and the cost of

Table 2: Logical Form Mutations

Mutation Type	Description
Predicate Addition	Introduces a new predicate symbol to the language.
Constant Addition	Introduces a new constant symbol to the formal system.
Variable Addition	Adds a new variable to the set of arbitrary objects.
Constant-to-Variable Substitution	Replaces constants with $\forall$ or $\exists$ quantified variables.
Conjunctive Atom Insertion	Conjoins new atomic formulas to existing states.
Disjunctive State Addition	Disjunctively creates new states with atoms.
Atom Negation	Applies or removes negation from atomic formulas.

re-running all model evaluations, we report results on this vetted 383-set; the generation pipeline supports regenerating larger sets for future work.

3.3 Mapping to natural language

To evaluate logical reasoning in language models in a more naturalistic context, we developed a systematic approach to convert formal logical statements into natural language prompts. We created 12 themes for natural language framings of logical problems, such as a researcher figuring out the properties of a novel element or a newly discovered creature. For each theme, we established consistent mappings between logical elements and thematic equivalents. The thematic conversion process serves multiple purposes: it ensures our results are robust against content effects [26]; it enables us to test whether the reasoning patterns are consistent in diverse contexts; and it mitigates potential data contamination, a significant problem for LLM evaluation [32–35] by reformulating problems in novel scenarios unlikely to appear in training data.

Consider the following example of a disjunction fallacy problem in its formal representation: $\mathbf{P1}$: $\{Q(x)R(y), S(x)T(y)\}, \mathbf{P2} : \{Q(x)\}$. In the "alchemy" theme, logical predicates (e.g., $A(x)$, $B(y)$) were mapped to thematic attributes (e.g., "is transmuting", "is time-bending"), while logical variables (e.g., x, y) were mapped to imaginary thematic entities (e.g., "cosmic dust", "vital mercury"). Using our alchemy theme, this was transformed into:

```
I'm an alchemist studying mysterious substances in my laboratory. I need to
     understand their properties through logical reasoning. Here's what I'
    ve discovered:

Cosmic dust is transmuting and vital mercury is time-bending, or cosmic
    dust is immortality-granting and vital mercury is spirit-affecting.
Also, we know that cosmic dust is immortality-granting.

What if anything follows? I do not have an intended answer in mind, and it
    is possible that nothing follows. Please be succinct and precise.
```

This is typical of the prompts sent to the LLM. It includes a thematic preamble and a natural language prompt to reason in an open-ended fashion with: "What if anything follows?" Each problem maintained the same prompt structure: an introduction establishing the thematic context, followed by the premises expressed in natural language, and concluding with a standardised question asking what follows from the premises. The prompt explicitly stated that nothing might follow, to avoid biasing models toward making unwarranted inferences [27].

3.4 Data Collection

Model Selection and Testing Conditions We evaluated 38 language models of varying sophistication, including several specifically designed as reasoning models. Selection informally prioritized quantity, popularity of usage, and representation of a diverse range of scores. All models were accessed via OpenRouter and tested using LLMharness.

Chatbot Arena scores were obtained from `https://huggingface.co/spaces/lmarena-ai/chatbot-arena-leaderboard/resolve/main/elo_results_20250423.pkl`. Scores for the HELM Capabilities benchmark for 9 models were obtained from `https://crfm.stanford.edu/helm/capabilities/v1.10.0/#/leaderboard` [17] and estimates for training compute for 19 models were obtained from `https://epoch.ai/data/ai-models` [16] (both URLs accessed 30 July 2025).

To ensure a fair comparison across architectural differences, all reasoning models were allocated 2,400 thinking tokens during evaluation. This standardization was justified given our use of the Chatbot Arena Elo score as our primary capability metric, which already accounts for performance differences regardless of underlying mechanisms [14].

Response Processing and Filtering Models provided conclusions in natural language, which we converted to PyETR format for evaluation. To ensure fair assessment of reasoning (not formatting), GPT-4.1-mini served as our translation layer without access to the original premises. Spot-checking confirmed translation fidelity.

Each model was evaluated on 383 reasoning tasks generated by PyETR. During analysis, we encountered occasional parse errors when processing model responses. To maintain data quality while preserving statistical power, models with parse error rates exceeding 20% across the test set were excluded from analysis, which resulted in the exclusion of one model (`google__gemini-2.5-pro-preview-03-25`). For the remaining models, individual responses that produced parse errors were excluded from analysis on a case-by-case basis rather than discarding all responses from that model.

Testing Framework The full set of logical problems were framed as evals, using Eleuther's Language Model Evaluation Harness [36], which handles evaling and metric collection against a wide range of models. OpenRouter [37] was used for all models, in order to have a unified API. This consumed less than $1,000 of compute resources. Each model was given a token limit of 3,000 output tokens.

Key Measures We collected the following measures to evaluate both the logical correctness of model responses and their alignment with human-like reasoning patterns, enabling our analysis of how these patterns correlate with model sophistication. A model's answer was considered **'logically correct'** if it was a logical consequence of the premises. This was tested using PySMT [38] to check whether the negation of the conclusion was inconsistent with the premises. LLM responses were classified as **'ETR-predicted'** if ETR predicts an endorsement of that conclusion according to PyETR's implementation of `default_procedure_does_it_follow`.

Statistical Tests We selected Pearson correlation to capture linear relationships between variables while also reporting Spearman's rank correlation to account for potentially non-linear monotonic relationships without assuming normality in the distribution of fallacy rates. For our premise-order intervention analysis, we employed two-proportion z-tests to rigorously evaluate whether reversing premise order produced statistically significant changes in fallacy rates. The z-test was specifically chosen because it allows us to determine if the observed differences in proportions (original vs. reversed premise order) represent genuine effects rather than random variation, providing a reliable measure of the significance of order effects across different model capabilities. This statistical approach enables us to quantify the extent to which models exhibit the same premise-order sensitivity documented in human reasoning studies.

4 Experimental Results

Our analysis reveals several notable patterns in how language models exhibit human-like reasoning fallacies. We report the results of Pearson correlation tests, as well as Spearman tests for the detection of non-parametric monotonic relationships. We consider $p \leq 0.05$ a suitable boundary for statistical significance.

Proportion of Errors that are Fallacies We operationalise a "human-like fallacy" (or simply "fallacy") as a response that is both ETR-predicted and logically incorrect. This definition captures